A QUASI-FINITE FINITELY PRESENTED MORPHISM OF AFFINE SCHEMES IS DECOMPOSED BY A FINITE LOCALLY FREE COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the parenthetical '(Details omitted.)' at source line 12114 concerns the standard descent of an algebra along a filtered colimit, whose inputs are named, and the remaining reductions are written out.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Descent of quasi-finiteness. Limits, Lemma limits-lemma-descend-quasi-finite, together with Algebra, Lemma algebra-lemma-colimit-category-fp-algebras. Reduces the statement to finite type algebras over the integers, that is, to the Noetherian case.
- Generic finiteness and Noetherian induction. Same chapter, Lemma lemma-generically-finite. Produces the partition into strata on which the morphism becomes finite.
- Structure of a finite morphism and base change of finite locally free morphisms. Morphisms, Lemmas morphisms-lemma-massage-finite, morphisms-lemma-finite-flat and morphisms-lemma-composition-finite-locally-free. Supplies the model in which the finite morphism is a finite union of closed subschemes mapping isomorphically, and keeps finite local freeness under composition.

Source reference. Original lemma 05N9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12110–12110. Navigation pointer invoked at proof

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line 12110; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 094M: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12112–12112. Navigation pointer invoked at proof line 12112; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03S1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12069–12069. Navigation pointer invoked at proof line 12118; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02KB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12136–12136. Navigation pointer invoked at proof line 12136; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02KC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12139–12139. Navigation pointer invoked at proof line 12139; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03HW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12146–12146. Navigation pointer invoked at proof line 12146; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-finite and finitely presented morphism of affine schemes.*

1. *There exists a surjective morphism of affine schemes $X' \to X$ and a closed subscheme $Z' \subset Y' = X' \times_X Y$ such that*
   (a) *$Z' \subset Y'$ is a thickening, and*
   (b) *$Z' \to X'$ is a finite étale morphism.*
2. *There exists a finite partition $X = \coprod X_i$ by locally closed, constructible, affine strata, and surjective finite locally free morphisms $X_i' \to X_i$ such that the reduction of $Y_i' = X_i' \times_X Y \to X_i'$ is isomorphic to $\coprod_{j=1}^{n_i} (X_i')_{red} \to (X_i')_{red}$ for some n_i.*

Proof. Setting $X' = \coprod X_i'$ we see that (2) implies (1). Write $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. Write A as a filtered colimit of finite type $\mathbf{Z}$-algebras A_i. Since B is an A-algebra of finite presentation, we see that there exists $0 \in I$ and a finite type ring map $A_0 \to B_0$ such that $B = \mathrm{colim}\, B_i$ with $B_i = A_i \otimes_{A_0} B_0$, see Algebra, Lemma 05N9. For i sufficiently large we see that $A_i \to B_i$ is quasi-finite, see Limits, Lemma 094M. Thus we reduce to the case of finite type algebras over $\mathbf{Z}$, in particular we reduce to the Noetherian case. (Details omitted.)

Assume X and Y Noetherian. In this case any locally closed subset of X is constructible. By Lemma 03S1 and Noetherian induction we see that there is a finite partition $X = \coprod X_i$ of X by locally closed strata such that $Y \times_X X_i \to X_i$ is

finite. We can refine this partition to get affine strata. Thus after replacing X by $X' = \coprod X_i$ we may assume $Y \to X$ is finite.

Assume X and Y Noetherian and $Y \to X$ finite. Suppose that we can prove (2) after base change by a surjective, flat, quasi-finite morphism $U \to X$. Thus we have a partition $U = \coprod U_i$ and finite locally free morphisms $U_i' \to U_i$ such that $U_i' \times_X Y \to U_i'$ is isomorphic to $\coprod_{j=1}^{n_i} (U_i')_{red} \to (U_i')_{red}$ for some n_i. Then, by the argument in the previous paragraph, we can find a partition $X = \coprod X_j$ with locally closed affine strata such that $X_j \times_X U_i \to X_j$ is finite for all i, j. By Morphisms, Lemma 02KB each $X_j \times_X U_i \to X_j$ is finite locally free. Hence $X_j \times_X U_i' \to X_j$ is finite locally free (Morphisms, Lemma 02KC). It follows that $X = \coprod X_j$ and $X_j' = \coprod_i X_j \times_X U_i'$ is a solution for $Y \to X$. Thus it suffices to prove the result (in the Noetherian case) after a surjective flat quasi-finite base change.

Applying Morphisms, Lemma 03HW we see we may assume that Y is a closed subscheme of an affine scheme Z which is (set theoretically) a finite union $Z = \bigcup_{i \in I} Z_i$ of closed subschemes mapping isomorphically to X. In this case we will find a finite partition of $X = \coprod X_j$ with affine locally closed strata that works (in other words $X_j' = X_j$). Set $T_i = Y \cap Z_i$. This is a closed subscheme of X. As X is Noetherian we can find a finite partition of $X = \coprod X_j$ by affine locally closed subschemes, such that each $X_j \times_X T_i$ is (set theoretically) a union of strata $X_j \times_X Z_i$. Replacing X by X_j we see that we may assume $I = I_1 \amalg I_2$ with $Z_i \subset Y$ for $i \in I_1$ and $Z_i \cap Y = \emptyset$ for $i \in I_2$. Replacing Z by $\bigcup_{i \in I_1} Z_i$ we see that we may assume $Y = Z$. Finally, we can replace X again by the members of a partition as above such that for every $i, i' \subset I$ the intersection $Z_i \cap Z_{i'}$ is either empty or (set theoretically) equal to Z_i and $Z_{i'}$. This clearly means that Y is (set theoretically) equal to a disjoint union of the Z_i which is what we wanted to show. $\square$